

● MEDIUM

● **ADVANCED MATH**

---

# Advanced Math

# 04

10 questions · ~15 min

Q1

GRID-IN ADVANCED MATH

The exponential function  $g$  is defined by  $g(x) = 19 \cdot a^x$ , where  $a$  is a positive constant. If  $g(3) = 2,375$ , what is the value of  $g(4)$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

ADVANCED MATH

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers  $b$ ,  $x$ , and  $y$ . Which equation correctly expresses  $x$  in terms of  $b$  and  $y$ ?

A.  $x = \frac{b - 72}{y}$

B.  $x = by - 72$

C.  $x = \frac{by - 72}{y}$

D.  $x = by - 72y$

Q3

ADVANCED MATH

$$(x - 11y)(2x - 3y) - 12y(-2x + 3y)$$

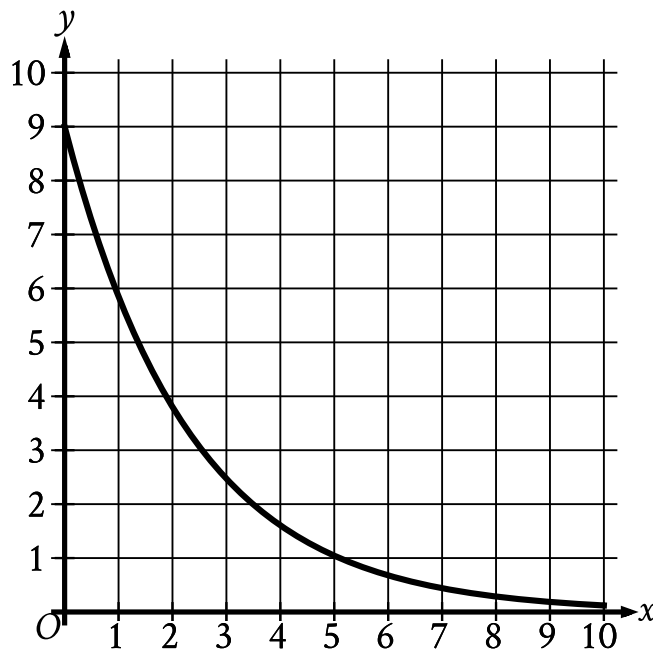
Which of the following is equivalent to the expression above?

A.  $x - 23y$

B.  $2x^2 - xy - 3y^2$

C.  $2x^2 + 24xy + 36y^2$

D.  $2x^2 - 49xy + 69y^2$



- Moving from left to right:
  - The curve is in quadrant 1.
  - From point (0 comma 9), the curve trends down sharply.
  - From point (5 comma 1), the curve trends down gradually.

The graph gives the estimated number of catalogs  $y$ , in thousands, a company sent to its customers at the end of each year, where  $x$  represents the number of years since the end of 1992, where  $0 \leq x \leq 10$ . Which statement is the best interpretation of the  $y$ -intercept in this context?

- A. The estimated total number of catalogs the company sent to its customers during the first 10 years was 9,000.
- B. The estimated total number of catalogs the company sent to its customers from the end of 1992 to the end of 2002 was 90.

- C. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9.
- D. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9,000.

Q5

ADVANCED MATH

$$v^2 = \frac{LT}{m}$$

The formula above expresses the square of the speed  $v$  of a wave moving along a string in terms of tension  $T$ , mass  $m$ , and length  $L$  of the string. What is  $T$  in terms of  $m$ ,  $v$ , and  $L$ ?

- A.  $T = \frac{mv^2}{L}$
- B.  $T = \frac{m}{v^2L}$
- C.  $T = \frac{mL}{v^2}$
- D.  $T = \frac{L}{mv^2}$

$$(2x+3)-(x-7)$$

Which of the following is equivalent to the given expression?

- A.  $x - 4$
- B.  $3x - 4$
- C.  $x + 10$
- D.  $2x^2 + 21$

| $x$ | $fx$ |
|-----|------|
| -1  | 10   |
| 0   | 14   |
| 1   | 20   |

For the quadratic function  $f$ , the table shows three values of  $x$  and their corresponding values of  $fx$ . Which equation defines  $f$ ?

- A.  $fx = 3x^2 + 3x + 14$
- B.  $fx = 5x^2 + x + 14$
- C.  $fx = 9x^2 - x + 14$
- D.  $fx = x^2 + 5x + 14$

$$y = 4x$$

$$y = x^2 - 12$$

A solution to the given system of equations is  $x, y$ , where  $x > 0$ . What is the value of  $x$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |
| 5                    | 5                    | 5                    | 5                    |
| 6                    | 6                    | 6                    | 6                    |
| 7                    | 7                    | 7                    | 7                    |
| 8                    | 8                    | 8                    | 8                    |
| 9                    | 9                    | 9                    | 9                    |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



**Q9**

ADVANCED MATH

$$x^2 + 6x + 4$$

Which of the following is equivalent to the expression above?

- A.  $(x + 3)^2 + 5$
- B.  $(x + 3)^2 - 5$
- C.  $(x - 3)^2 + 5$
- D.  $(x - 3)^2 - 5$

**Q10**

ADVANCED MATH

The area  $A$ , in square centimeters, of a rectangular cutting board can be represented by the expression  $ww + 9$ , where  $w$  is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

- A.  $ww + 9$
- B.  $w$
- C.  $9$
- D.  $w + 9$